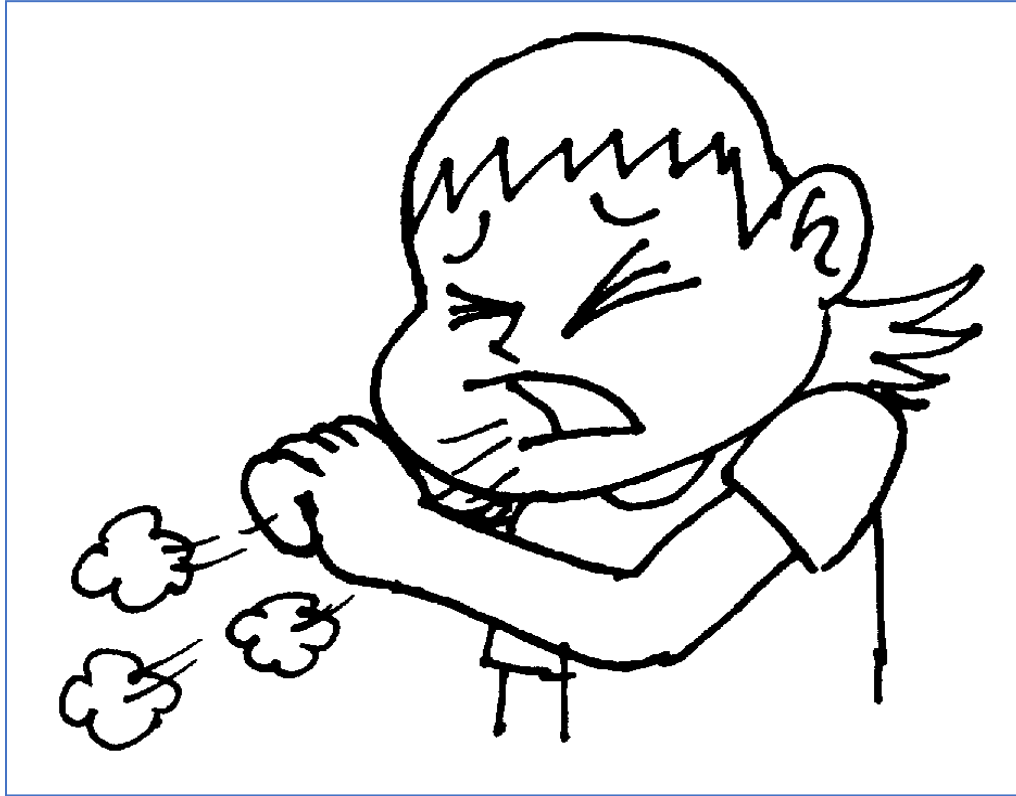


Pharmacotherapy of cough



By;

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Introduction

- ❖ Coughing is a protective reflex action caused when the airway is being irritated or obstructed.
- ❖ Its purpose is to clear the airway so that breathing can continue normally.
- ❖ The majority of coughs caused by upper respiratory viral infection.

- ❖ Cough is a useful physiological mechanism that serves to clear the respiratory passages of foreign material and excess secretions.
- ❖ It should **not be suppressed indiscriminately**.
- ❖ There are, however, many situations in which cough does not serve any useful purpose but may, instead only **annoy the patient or prevent rest and sleep**.
- ❖ Chronic cough can contribute to **fatigue**, especially in elderly patients, in such situations **we should use a drug that will reduce the frequency or intensity of the coughing**.

- ❖ Cough reflex is complex, involving the central and peripheral nervous systems as well as the smooth muscle of the bronchial tree.
- ❖ It has been suggested that irritation of the bronchial mucosa causes bronchoconstriction, which in turn, stimulates cough receptors (which probably represent a specialized type of stretch receptor) located in the tracheobronchial passages.

Mechanism of cough

Stimulation of mechano-or chemoreceptors (throat, respiratory passages or stretch receptors in lungs)



Afferent impulses to cough centre (medulla)



Efferent impulses via parasympathetic & motor nerves to diaphragm, intercostal muscles & lung



Increased contraction of diaphragmatic, abdominal & intercostal (ribs) muscles $\Rightarrow$ **noisy expiration (cough)**

Types of cough

1.Acute Cough: Lasts less than 3 weeks (e.g., viral infections).

2.Subacute Cough: Lasts 3-8 weeks (e.g., post-infectious).

3.Chronic Cough: Lasts more than 8 weeks (e.g., asthma, GERD).

□Cough may be

i) Un productive (dry) cough **OR**

ii) Productive cough (sputum)

Most common causes of cough

- Upper/lower respiratory tract infection (Viral (cold, flu), bacterial (pneumonia, TB))
- Allergic rhinitis
- Smoking
- Chronic bronchitis
- Asthma
- Gastroesophageal reflux
- Congestive heart failure
- Bronchiectasis
- Use of drugs (e.g., Angiotensin-converting enzyme inhibitors)

Red Flag Symptoms with Report of Nonspecific Cough or Other Respiratory Symptoms

- Persistent cough lasting more than 8 weeks.
- Cough with blood (hemoptysis).
- Shortness of breath or wheezing.
- Fever, weight loss, or night sweats.

Red Flag Symptoms with Report of Nonspecific Cough or Other Respiratory Symptoms

- ❖ Hemoptysis
- ❖ Smoker >45 years of age with a new cough, change in cough, or coexisting voice disturbance
- ❖ Adults aged 55-80 years who have a 30 pack-year smoking history and currently smoke or who have quit within the past 15 years
- ❖ Prominent dyspnea, especially at rest or at night
- ❖ Hoarseness
- ❖ Systemic symptoms (ie, fever, weight loss, peripheral edema with weight gain)
- ❖ Trouble swallowing when eating or drinking
- ❖ Vomiting
- ❖ Recurrent pneumonia
- ❖ Abnormal respiratory exam and/or abnormal chest radiograph coinciding with duration of cough

Diagnosis of Cough

- **Medical History & Physical Examination.**
- **Chest X-ray or CT scan for chronic cases.**
- **Lung Function Tests (Spirometry for asthma/COPD).**
- **Allergy Testing.**
- **Bronchoscopy (for persistent cases).**

Treatment & Management

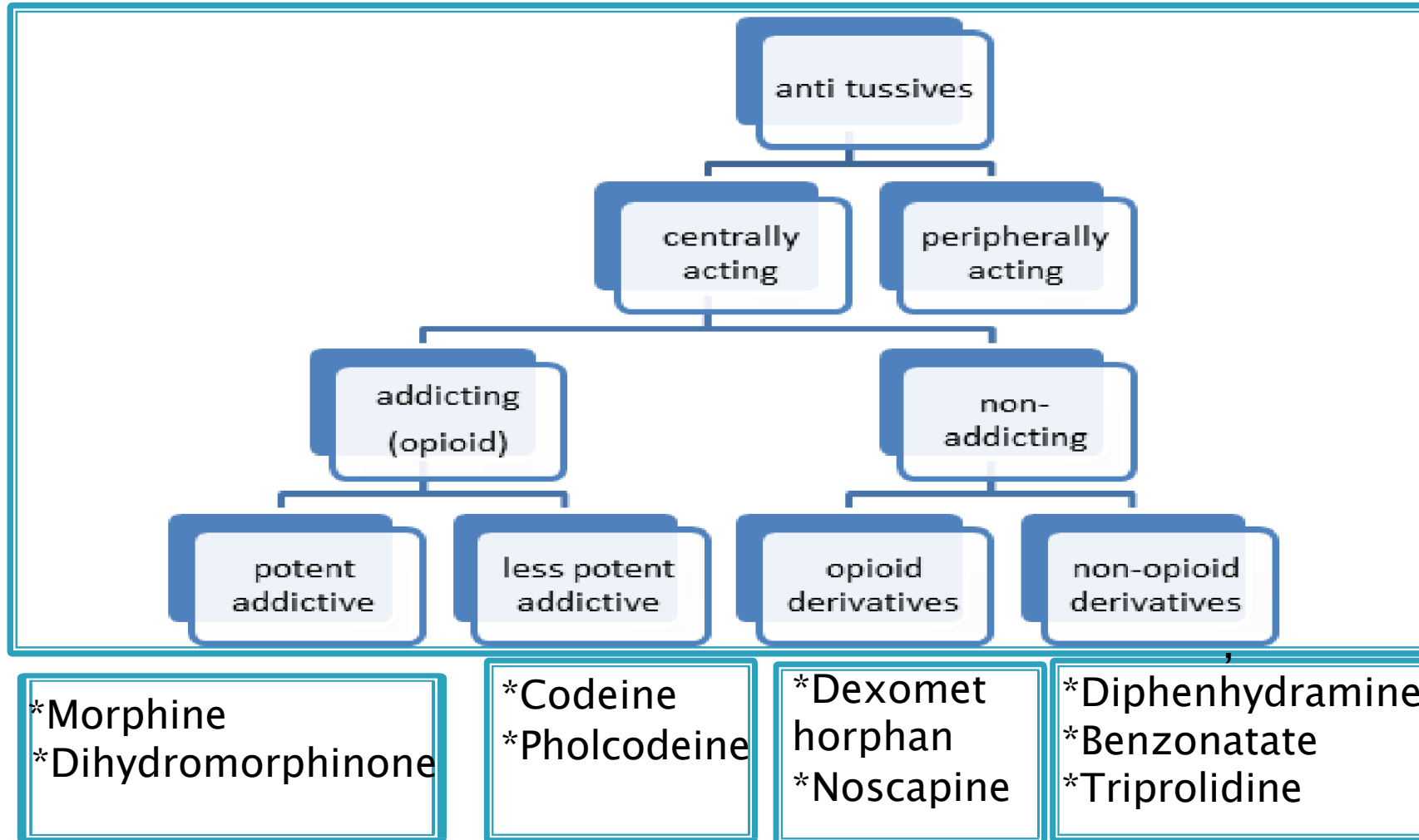
- 1.Lifestyle Modifications:** Avoid smoking, use humidifiers.
- 2.Medications:** Cough suppressants, expectorants, bronchodilators.
- 3.Treating Underlying Causes:** Asthma inhalers, GERD medication.
- 4.Home Remedies:** Honey, steam inhalation, hydration.

Treatment of Cough



- 1) **Antitussives** (cough centre suppressants)
- 2) **Expectorants (Mucokinetics)**
- 3) **Mucolytics**
- 4) **Antihistamines**
- 5) **Bronchodilators**
- 6) **Pharyngeal Demulcents**

ANTI TUSSIVE AGENTS



1) Antitussives (cough centre suppressants)

**Drugs suppress cough & produces symptomatic relief
For unproductive cough**

MOA

Mainly suppress cough centre in medulla (both central & peripheral effects)

E.g., **Opioid drugs** (codeine, pholcodeine, noscapine, dextromethorphan)

Opioid = most effective for cough

i) Codeine

- Codeine= pro drug $\Rightarrow$ metabolized to morphine
- ✓ It is an opium alkaloid
- ✓ Has less addiction
- ✓ Are resp. centre depressant
- ✓ Has useful antitussive action at low doses (<15 mg)

Adverse Effects

- Constipation
- thickening of sputum
- Respiratory depression & drowsiness

ii) Noscapine & Pholcodeine

- Related to papaverine
- Do not have addictive, analgesic & constipating properties
- Do not interfere with mucocilliary movement
- Noscapine (15 mg) & pholcodeine (10 mg)=syrup

Adverse Effects

- Nausea
- Drowsiness
- Headache

iii) Dextromethorphan

➤ Available in syrup, tablets, spray forms

MOA

NMDA receptor antagonist

Uses

Cough suppressant, temporary relief of cough caused by **minor throat & bronchial irritation** (accompanies with flu & cold), pain relief

Adv. Effects= Nausea, vomiting, drowsiness, dizziness, blurred vision

2) Expectorants (Mucokinetics)

- Act peripherally
- Increase bronchial secretion
- OR
- Decrease its viscosity $\Rightarrow$ facilitates its removal by coughing
- Loose cough $\blacktriangleright$ less tiring & more productive
- ❑ Classified into
 - a) **Directly acting**
E.g., Guaifenesin (glyceryl guaiacolate), Na^+ & K^+ citrate or acetate,
 - b) **Reflexly acting**
E.g., Ammonium salt

i) Sodium & potassium citrate or Acetate

- **They act directly**

Actions: increase Bronchial secretion by salt action

ii) Guaifenesin

- Expectorant drug usually taken by mouth
- Available as single & also in combination

MOA=Increase the volume & reduce the viscosity of secretion in trachea & bronchi

iii) Reflexly acting

Ammonium salts

Gastric irritants $\Rightarrow$ reflexly $\uparrow$ bronchial secretions + sweating

3) Mucolytics

Help in expectoration by **liquefy** the viscous tracheobronchial secretions

E.g., ***Bromhexine, Acetyl cysteine, Carbocysteine, Methylcysteine.***

i) Bromhexine

Synthetic derivative of **vasicine** (alkaloid= *Adhatoda vasica*)

MOA of Bromhexine

- a) Thinning & fragmentation of mucopolysaccharide fibers
- b) ↑ volume & ↓ viscosity of sputum

ii) Acetylcysteine

Given directly into respiratory tract

MOA of acetylcysteine

Opens **disulfide bond** in mucoproteins of sputum = ↓ viscosity

Uses

Cystic fibrosis (to ↓ viscosity of sputum)

Onset of action quick---used 2-8 hourly

Adverse effects

Nausea, vomiting, bronchospasm in bronchial asthma

4) Antihistamines

- Added to antitussives/expectorant formulation
- Due to sedative & anticholinergic actions produce relief in cough **but lack selectivity for cough centre**
- **Non sedating antihistamine are not effective**
- No expectorant action = decrease secretions (anticholinergic effect)
- Suitable for allergic cough (not for asthma)
E.g., ***Chlorpheniramine, diphenhydramine, promethazine***

5) Bronchodilators

- ❑ Bronchospasm or stimulation of pulmonary receptors = induce or aggravate cough + bronchoconstriction
- ❑ e.g. β_2 -agonist (salbutamol, terbutaline)

MOA of bronchodilators in cough

- ❑ Increase surface velocity of air flow during cough → Clear secretions of airway
- ❑ Not used routinely for every type of cough but only when bronchoconstriction is present

6) Pharyngeal demulcents

- Soothe the throat (directly & also by promoting salivation)

- ❖ Decrease afferent impulses from inflamed/irritated pharyngeal mucosa
- ❖ Provide symptomatic relief in dry cough arising from throat

E.g. lozenges, cough drops, glycerine, liquorice, honey

Specific treatment approach to cough

Etiology of cough

Treatment

- | | |
|--|--|
| 1) Upper/lower respiratory tract infections | Appropriate antibiotics |
| 2) Smoking/chronic bronchitis | Cessation of smoking |
| 3) Pulmonary tuberculosis | Antibiotics |
| 4) Asthmatic cough | Inhaled β2-agonists/ipratropium/corticosteroid |
| 5) Postnasal drip (sinusitis) | Antibiotics, nasal decongestants, antihistamines |

First Line medication for cough

- ❑ 1st-generation antihistamine should be empiric treatment:
 - Dexbrompheniramine 6 mg/pseudoephedrine 120 mg PO q.12 h: Age >12 years.
- ❑ For short-term symptomatic relief of nonproductive cough:
 - Dextromethorphan 10–20 mg PO q.4h: Age >12 years. Use 5–10 mg PO q.4h for age 6–12 years.
 - Narcotics: Codeine 15–30 mg PO q.6h; hydrocodone 5 mg PO q.6h

Prevention Strategies

- **Hand Hygiene** to prevent infections.
- **Vaccination** (Flu, Pneumonia).
- **Avoiding known Irritants** (Smoke, Pollution).

THANKS